

VoltGrid — Slide 3 Flowchart & System Architecture Guide

SIH 2026 Presentation Template — Slide 3: System Flow Chart / Process & Methodology.

Reference Model: Inspired by the SIH 2025 Finalist Architecture Flowchart (ZyraNav / Helix House).

Platform: VoltGrid — Intelligent EV Energy & Route Optimization Platform.

1. Flowchart Architecture Concept (The 4 Functional Blocks)

In the reference SIH 2025 finalist slide, the flowchart succeeded because it avoided a single straight line. Instead, it showed:

1. **The Primary Core Flow** (Trip request \rightarrow Road geometry \rightarrow Energy math \rightarrow Feasibility check).
2. **The Adaptive Optimization Block** (Multi-stop forward convergence, payload & aerodynamic drag adjustments).
3. **The Predictive ML & Scoring Loop** (Logistic regression vacancy model at ETA, venue profiles, queue forecasting).
4. **The Failover & Autonomous Recovery Branches** (In-transit outage reroute in <500 ms, overnight hotel stays, emergency 15A swapping kiosks).

VOLTGRID SYSTEM FLOW CHART



2. Mermaid Diagram Code

(Copy and paste this into Mermaid Live Editor or your markdown viewer)

```
graph TD
    INP["1. TRIP REQUEST INPUT<br>Origin, Destination, EV Model (26 presets)<br>Starting SoC%, Payload Mass, Preference"] --> SEC["2. API SECURITY BOUNDARY<br>Zod Strict Validation • 128 KiB Payload Cap<br>Token-Bucket IP Rate Limiting (120 req/min)"]
    SEC --> OSRM["3. OSRM DRIVING ENGINE<br>Pairwise Road Geometry • Highway Distance (km)<br>10s Timeout with Haversine Fallback"]
    OSRM --> DEC1["4. DIRECT FEASIBLE?<br>Can arrive with ≥ desired<br>arrival battery reserve?"]
    DEC1 -- "YES (Arrival SoC ≥ Reserve)" --> DIR["DIRECT 0-STOP ROUTE<br>Continuous highway polyline<br>Guaranteed battery buffer"]
    DEC1 -- "NO (Charging Required)" --> OPT["5. CORRIDOR OPTIMIZATION ENGINE<br>Spatial Bounding Box (Corridor Filter)<br>findMultiStop() Forward Directional Progress"]
    OPT --> PHY["6. FIRST-PRINCIPLES PHYSICS CORE<br>Wh/km = Base × M_occupancy × M_terrain × M_weather + P_cargo<br>Pillion Wind Drag (+)"]
    PHY --> ML["7. PREDICTIVE AVAILABILITY ML<br>Logistic Regression: P(avail at ETA) = σ(w·x + b)<br>Venue Profile Demand Curves (highway)"]
    ML --> SCORE["8. MULTI-ATTRIBUTE SCORER<br>Score = w_avail·P_avail - w_detour·D - w_time·T<br>+ w_rel·R - w_cost·C"]
    SCORE --> DEC2["9. IN-TRANSIT OUTAGE?<br>Charger offline or tripped?"]
    DEC2 -- "NO (Normal Operation)" --> OUT["10. MULTI-STAKEHOLDER OUTPUT<br>• Driver: Interactive Leaflet Map with Leg-by-Leg Recharge Delta Bars<br>• Reservation"]
    DEC2 -- "YES (Charger Outage Detected)" --> REROUTE["AUTONOMOUS RE-ROUTING<br>/api/reroute in <500 ms<br>Instant diversion to backup CPO hub"]
    REROUTE --> OUT

    %% Connections
    INP --> SEC
    SEC --> OSRM
    OSRM --> DEC1
    DEC1 -- "YES (Arrival SoC ≥ Reserve)" --> DIR
    DEC1 -- "NO (Charging Required)" --> OPT
    OPT --> PHY
    PHY --> ML
    ML --> SCORE
    SCORE --> DEC2
    DEC2 -- "NO (Normal Operation)" --> OUT
    DEC2 -- "YES (Charger Outage Detected)" --> REROUTE
    REROUTE --> OUT
    DIR --> OUT
```

3. How to Draw This on Slide 3 (Canva / PowerPoint / Figma Instructions)

To recreate the look of the **SIH 2025 Finalist (ZyraNav)** slide:

1. Left Side (30% width):

- Keep your **Tech Stack & Proof of Work** (Next.js 16, OSRM, Leaflet, PostGIS, 54 unit tests, 925+ stations, GitHub link).

1. Right Side (70% width) — FLOW CHART:

- Box 1 (Top Blue):** **TRIP INPUT & SECURITY** (Inputs → Zod validation → OSRM engine).
- Central Decision Diamond (Yellow):** **DIRECT FEASIBLE?**
- Left Arrow (Green): **0-Stop Direct Route**.
- Right Arrow: Leads into the **Corridor Charging Pipeline**.
- Box 2 (Emerald Green):** **FIRST-PRINCIPLES PHYSICS CORE** (Wh/km, Pillion Drag +15%, Cargo mass, LFP/NMC chemistry).
- Box 3 (Purple/Lilac):** **PREDICTIVE AVAILABILITY ML ENGINE** (Logistic Regression $\sigma(\mathbf{w} \cdot \mathbf{x} + \mathbf{b})$, venue profiles, queue forecasting).
- Box 4 (Orange):** **MULTI-ATTRIBUTE SCORING** (Detour, Power, Availability, Cost, Reliability).

- **Sidecar Alert Box (Red/Pink):** **AUTONOMOUS RE-ROUTING (<500ms)** showing how outages trigger instant in-flight recovery.
- **Bottom Box (Dark Navy):** **MULTI-STAKEHOLDER OUTPUT** (Driver map with delta bars, Razorpay QR booking pass, Operator grid HUD).

4. The 30-Second Flowchart Pitch for Judges

When you point to this diagram during the evaluation:

*"Judges, looking at our central flowchart:

When a user enters their trip, it doesn't just ping a database. The request passes through our **Zod security gate**, and **OSRM** extracts the real highway road geometry.

At our first decision gate, we test if the destination is reachable directly. If not, our **Corridor Optimization Engine** activates two parallel brains:

1. Our **First-Principles Physics Core** calculates real-world Wh/km consumption, factoring in pillion aerodynamic drag, passenger weight, and battery chemistry.

2. Simultaneously, our **Logistic Regression ML Model** forecasts charger vacancy at the driver's estimated time of arrival.

Our multi-attribute scoring function selects the optimal stop. And notice our safety branch on the right: if an assigned charger trips while the car is in motion, our **autonomous rerouting engine** recalculates in under 500 milliseconds, outputting a guaranteed slot booking, a digital QR pass, and live operator telemetry."*